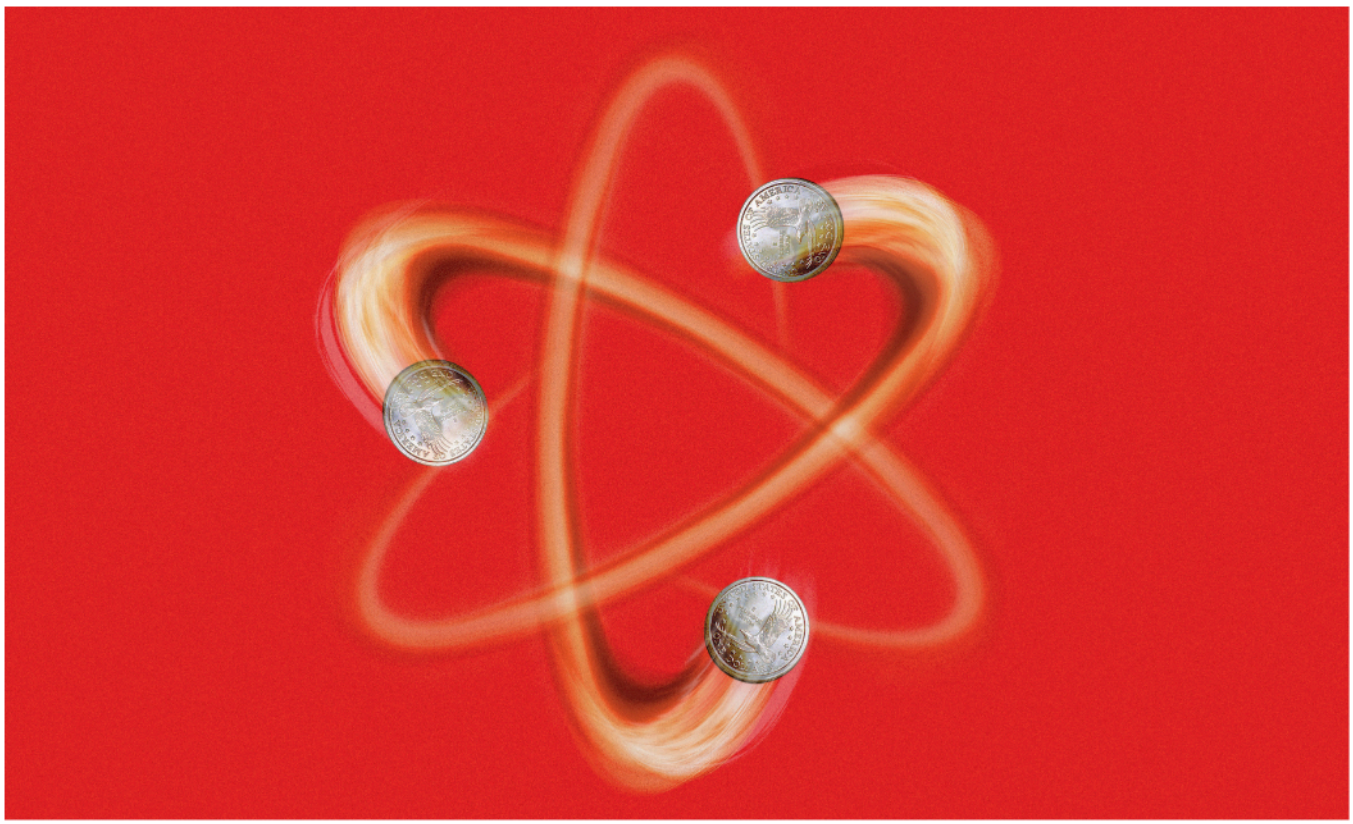
Science & technology



Science funding

Back to the bench

America is entering a new era of private science

THE YEARS since the second world war have been good to American science. The country's researchers sent men to the moon, invented the modern computer and led the sequencing of the human genome. Its government funded basic research in academia and national labs, which could then pass on findings with commercial promise to firms.

Now there are signs the innovation machine is breaking down. The system's critics fret that America's scientific infrastructure encourages groupthink, since a small number of committees distribute a large share of funding for basic research. They worry that universities are ill-suited to the interdisciplinary work that much of 21st-century science requires. And America is no longer an undisputed scientific superpower: China's research and development (R&D) spending has caught up with Uncle Sam's in purchasing-power terms, and is growing faster. In 2024 China overtook America in the *Nature* Index, a measure of publications in the most prestigious scientific journals.

Advocates of "metascience", the study of scientific discovery, have spent years pointing out the foibles of the system. Lately they have been putting their ideas into practice. Private institutions, many of which were founded within the past five years, say they are building the future of science. The history of American innovation suggests they may be right.

Private science is nothing new. Many of the most exciting American discoveries of the 20th century were not dreamt up on university campuses, but within corporate research institutes. Scientists at Bell Labs (the research arm of AT&T), Xerox PARC (the copy-machine maker's lab) and IBM Research came up with such marvels as the transistor, and won a handful of Nobel prizes in so doing.

Yet for all their successes, private labs failed to justify their upkeep. A paper published in 2004 by William Nordhaus, an economist at Yale University, estimated that, in the post-war period and across the whole economy, just 2.2% of value from innovation flowed to firms. In response, companies transitioned to "R&D by M&A," buying startups spun out from universities rather than funding their own labs.

Corporate labs' retreat led basic research to become much more reliant on government-backed efforts. In the view of private-science advocates, that is lamentable. The concentration of science funding means fewer innovative ideas can benefit society at large, says Caleb Watney of the Institute for Progress, a think-tank. To wit, a study of 100,000 National Institutes of Health (NIH) grant applications, conducted in 2025 by Pierre Azoulay of the Massachusetts Institute of Technology and Wesley Greenblatt of Harvard Medical School, found that projects which took out-of-consensus approaches were less likely to have

→ ALSO IN THIS SECTION

80 Japan's mission to Mars's moons
81 Redefining the second
82 A global cyber-security incident
82 Well Informed: Outrunning injuries

their funding renewed. And seeking funding is onerous. According to one survey, researchers spend about 44% of their time doing administrative tasks like writing proposals and reports.

To illustrate these issues, many cite the example of Katalin Kariko, a biologist whose work provided much of the science underpinning mRNA vaccines, such as those for covid-19. For decades funders, notably the NIH, rejected her proposals, considering them too risky. She happened to stay in science, but not many in her position would have, says Stuart Buck of the Good Science Project, a lobby group.

Such concerns have made their way from metascientists' lips to powerful ears. On July 21st the American government released a report arguing that universities should not be "the only home of America's most innovative scientific research". Instead, funding agencies should act more like venture capitalists, betting on a wide variety of institutions.

Sovereign science

As part of this push, the National Science Foundation (NSF) is to allocate $1.5bn to a first round of "X-labs", new organisations that "advance breakthrough science outside of traditional institutions". A bipartisan bill in Congress would create a similar initiative at the NIH, which has a much larger budget, notes Josh Harder, a Democratic representative and one of the bill's sponsors. Other changes would see federal agencies offer "golden tickets" to oddball projects that a committee would never back, or begin to give long-term money to individuals rather than projects.

Such ideas have discomfited scientists because of their source. As it attempts to spark "a new golden age" for American science, the Trump administration wants to fund it less. The White House's proposed budget for fiscal year 2027 suggests halving the NSF's budget. The worry is that the combination of less funding spread across more institutions will lead to university labs losing out.

Michael Kratsios, the White House science czar, says he "disagrees with that premise". He notes that the administration has suggested boosting overall R&D spending, particularly in defence. (Defence-department programmes have spurred innovations such as epipens, bug spray and self-driving cars.) Nor does he think the bottom line is the right way to measure policy. Science advocacy groups "have this extreme fixation on budgets", he says. "Academia just needs to get out of this zero-sum mindset."

But separately from such squabbling, many private science experiments have already begun. One of the most notable of these is Convergent Research, a Massachusetts-based nonprofit founded in 2021 to fund "focused research organisations" (FROs). Its grantees often want to create a specific tool rather than answer a purely academic question, say co-founders Anastasia Gamick and Adam Marblestone. Convergent gives researchers $20m-50m and between three and seven years to carry out their project. Afterwards, its staff will help them decide what to do with the results. One FRO creates sets of tools that labs and companies can use to study non-model organisms. Another produces an interface that uses ultrasound waves to read brain activity.

Another entrant to the world of private science is Episteme, a company launched in 2025 by Sam Altman of OpenAI and Louis Andre, then a little-known researcher. Rather than demanding pre-set targets or papers, Episteme hopes to give its lead researchers near-total freedom. While the select few it funds often have an approach in mind to solving their chosen problem, Episteme offers them the flexibility to adapt. "If an idea fails, which in science is something that happens very often, you should be able to pivot and try a different angle," says Mr Andre.

The approach is an appealing one. Charles Marcus, a physicist, says he left a tenured position at the University of Washington to work at Episteme because he felt it would enable him to work across disciplines without needing approval from staid bureaucrats. "We're not building academic departments in which nobody talks to each other," he says. Nor will he and other lab leads need to apply for grants or wait for months to hire junior researchers. And while publishing papers is encouraged, it is not mandated. The main constraint on their work is that it should represent a commercial opportunity.

Many important questions fall outside that scope and will have to remain within universities. And besides, not every attempt at private science has flourished. Arena BioWorks, a private biomedical institute founded in 2023, shut down after less than two years. But whereas Arena's funders reportedly expected quick payouts, Mr Andre says his backers, who have funded Episteme from their personal fortunes, are committed for the long haul. And Episteme, unlike Arena, is not betting on one specific area of research, either. "If a programme fails, it's okay. We can recycle it and repurpose the resources into something else," he reckons.

Between federal support, independent experiments and a coming wave of AI-fuelled cash, private-science initiatives are "seeing the stars align," says Mr Marblestone. Historians of science and technology argue the post-war system cemented American hegemony. In the bet on private science the stakes are high, both for boffins and for America. ■

Space exploration

Phobophiles

TOKYO

Japan heads to Mars's moons

ORBITERS, LANDERS and rovers have travelled to Mars, but none has returned. The Japan Aerospace Exploration Agency (JAXA) intends for that to change. On October 20th its Martian Moons eXploration (MMX) mission lifted off from Tanegashima, an island in southern Japan. The probe will spend the next year travelling to Mars, where it aims to land on Phobos, the larger of the planet's two moons, collect at least ten grams of rock and dust, and return the haul to Earth in 2031. Planetary scientists hope the mission will resolve a long-standing puzzle: how Mars's moons were formed—and, in the process, reveal much about the planet itself.

Phobos and Mars's other moon, Deimos, are small, respectively about 22km and 12km across, and look like lumpy potatoes. That might suggest they are asteroids that were captured by Mars's gravitational field after straying too close. Captured asteroids, however, usually have tilted, elliptical orbits, whereas both moons follow nearly circular paths around the planet's equator. That hints at another possible origin: a giant collision that threw debris into orbit that later gathered into moons.

JAXA has spent over a decade developing MMX, working with the space agencies of America, Europe and Australia. The mission builds on Japan's record of returning samples from space. Its Hayabusa mission collected small amounts of grains



Moonshot

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